Exam 1

Mathematical Models

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Chapter 2

2. Reconsider the whale problem of Exercise 1, but now look at the total number of whales. We will say that the whale population levels x and y are feasible provided that both x and y are nonnegative. We will say that the population levels x and y are *sustainable* provided that both of the growth rates dx/dt and dy/dt are nonnegative.

(a) Determine the population levels that are feasible, sustainable, and that maximize the total whale population $x + y$. Use the five-step method, and model as a constrained optimization problem.

Solution. While ensuring that the population levels are feasible and sustainable, we find the total number of whales to be $138{,}207 + 393{,}090 = 531{,}297$. We found this number by first plotting $W(x, y) = x + y$, where W is a function that just counts x blue and y fin whales, and ensuring we stay within the sustainable region, where $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are positive (see Figure 1). Then, we set $\frac{dx}{dt}$ and $\frac{dy}{dt}$ equal to zero, and solve for y and x respectively. This gave us two line equations, which we plotted in Desmos, where we found their intersection (see Figure 2). We know this intersection is the maximum because of our first plot. (Note: I could have also found the intersection by setting each linear equation to 0 and solving for both, but I thought the Desmos visual was more appealing.)

Chapter 3

6. Reconsider the facility location problem of Example 3.2, but now assume that the response time from point (x_0, y_0) to point (x_1, y_1) is proportional to the road travel distance $|x_1 - x_0| + |y_1 - y_0|$.

(a) Find the location that minimizes average response time. Use the five-step method, and model as a multivariable unconstrained optimization problem. Python Code

Solution. To find the optimal location that minimizes average response time, we used contour maps to iteratively get closer to the optimal response time given by $z(x, y)$. Through this process (see Figure 3), we find that the contour plots continuously approach $x = 1$, $y = 3$ (going any further than $x \in [0.999, 1.001]$, $y \in [2.999, 3.001]$ breaks the plot). Plugging these values in for our response time function, we get $z(1, 3) = 7.145$, which we find to be the best response time. (Also, I used the same random search method that they tried in the book, and as I constricted the random ranges for x and y closer to 1 and 3, $z(x, y)$ never went lower than 7.145.)

		MI	NY	CA	GA	Output
Textile Mill	1	430	550	680	700	105
	2	510	590	890	685	160
	3	395	425	910	450	85
Allocation		70	100	105	75	

Table 1: (3.7) Shipping cost per truckload in dollars

15. The Burningham textiles company has three textile mills located in the southern U.S. and four distribution centers located in Michigan, New York, California, and Georgia. The estimated yearly output from each mill, the allocation to each warehouse, and the shipping costs are tabulated in Table 3.7.

(a) Find the transportation plan that minimizes shipping costs. Use the five-step method, and solve as a linear programming problem using a computer.

Solution. Through using the Simplex LP Solver on Excel, we find a minimum shipping cost of \$198,200 with the shipment distribution as follows:

$$\mathrm{CA}_1 = 105, \quad \mathrm{MI}_2 = 70, \quad \mathrm{NY}_2 = 90, \quad \mathrm{NY}_3 = 10, \quad \mathrm{GA}_3 = 75,$$

and with the rest receiving no utilization. The Excel sheet calculation is available in Figure 4.

Chapter 4

5. In the whale problem of Example 4.2 we used a logistic model of population growth, where the growth rate of population P in the absence of interspecies competition is

$$g(P) = rP\left(1 - \frac{P}{K}\right).$$

In this problem, we will be using a more complex model,

$$g(P) = rP\left(\frac{P-c}{P+c}\right)\left(1 - \frac{P}{K}\right),$$

in which the parameter c represents a minimum viable population level below which the growth rate is negative. Assume that $\alpha = 10^{-8}$ and that the minimum viable population level is 3,000 for blue whales and 15,000 for fin whales.

(a) Can the two species of whales coexist? Use the five-step method, and model as a dynamical system in steady state.

Solution. The two species of whales can coexist. They will reach a stable equilibrium at $(3535, 399810)$ (see Figure 5).

(b) Sketch the vector field for this model. Classify each equilibrium point as stable or unstable.

Solution. The vector field (stream plot) is available in Figure 5, and each equilibrium is classified in Table 2.

(c) Assuming that there are currently 5,000 blue whales and 70,000 fin whales, what does this model predict about the future of the two populations?

Solution. It depends on the growth of each species. One concern could be that the fin whales have a much higher starting population, so they could populate quickly and drive the blue whales to extinction. However, if that does not happen, then the two species can coexist, as explained in part (a).

Blue Whales	**Finn Whales**	**Eigen 1**	**Eigen 2**	**Type**
0	0	−0.078	−0.051	Sink
3000	0	−0.078	0.025	Saddle
150000	0	−0.078	−0.047	Sink
0	400000	−0.070	−0.055	Sink
0	15000	−0.051	0.039	Saddle
3018	15012	0.023	0.039	Source
149513	15595	−0.047	0.039	Saddle
137698	392568	−0.070	−0.043	Sink
3535	399810	−0.070	0.021	Saddle

Table 2: Display of equilibrium points, their eigenvalues, and the associated type for each.

Plots & Code

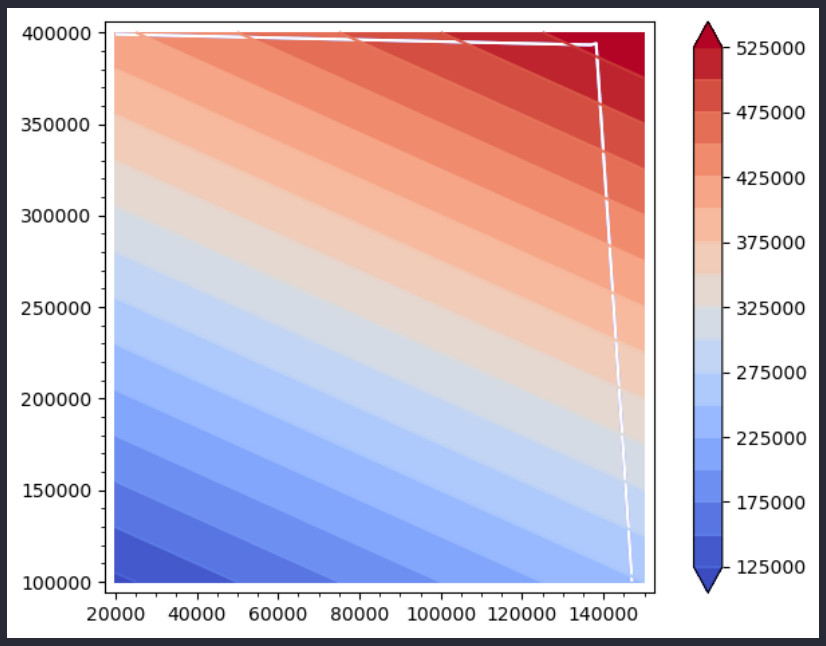


Figure 1: Contour plot for $W(x, y)$ with boundaries $\frac{dx}{dt} \geq 0$ and $\frac{dy}{dt} \geq 0$ (white lines) for Chapter 2, Exercise 2. The x-axis represents blue whales, and the y-axis represents fin whales.

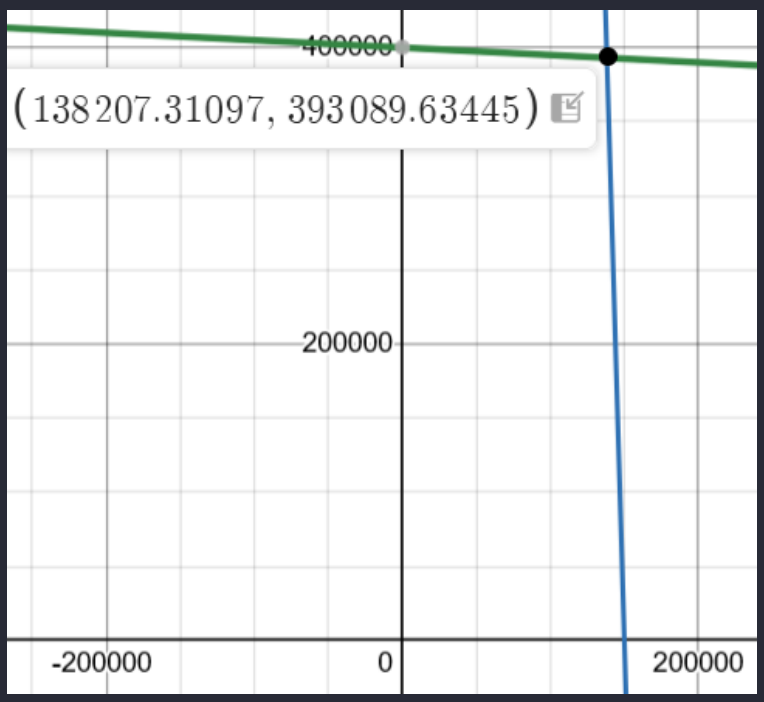
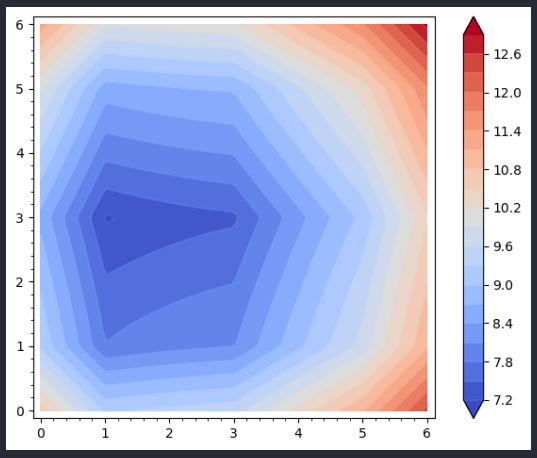
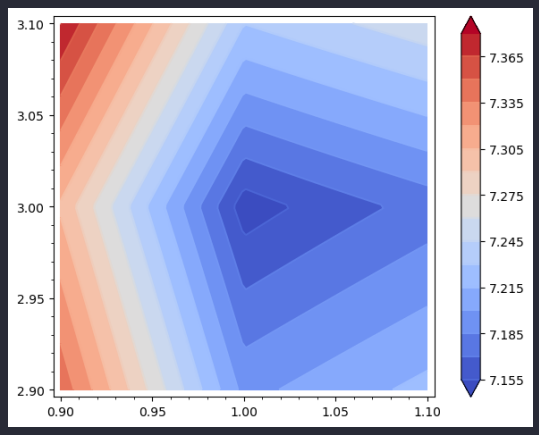


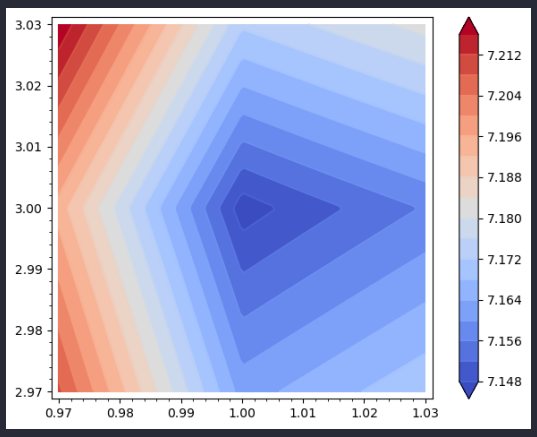
Figure 2: Intersection of the lines $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} = 0$ for Chapter 2, Exercise 2.



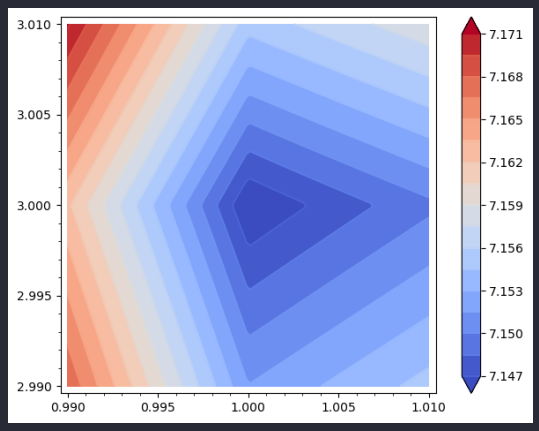
(a) 6×6 Contour Plot



(b) 1.1×3.1 Contour Plot



(c) 1.03×3.03 Contour Plot



(d) 1.01×3.01 Contour Plot

Figure 3: Contour plots for the response time function $z(x, y)$ in Chapter 3, Exercise 6. Degree of precision sequentially increases from (a) to (d).

```
x, y, z = var("x,y,z")
z(x,y) = 3.2 + 1.7 * (
    6 * (abs(x - 1) + abs(y - 5))^(0.91) + \
    8 * (abs(x - 3) + abs(y - 5))^(0.91) + \
    8 * (abs(x - 5) + abs(y - 5))^(0.91) + \
    21 * (abs(x - 1) + abs(y - 3))^(0.91) + \
    6 * (abs(x - 3) + abs(y - 3))^(0.91) + \
    3 * (abs(x - 5) + abs(y - 3))^(0.91) + \
    18 * (abs(x - 1) + abs(y - 1))^(0.91) + \
    8 * (abs(x - 3) + abs(y - 1))^(0.91) + \
    6 * (abs(x - 5) + abs(y - 1))^(0.91)
)/84
```

Listing 1: Response time function for Chapter 3, Exercise 6.

Textiles shipping minimization problem

	costs											
	1MI	1NY	1CA	1GA	2MI	2NY	2CA	2GA	3MI	3NY	3CA	3GA
	430	550	680	700	510	590	890	685	395	425	910	450

Inputs												
	1MI	1NY	1CA	1GA	2MI	2NY	2CA	2GA	3MI	3NY	3CA	3GA
	0	0	105	0	70	90	0	0	0	10	0	75

Constraint	Use	Lim	Slac
1	105	105	0
2	160	160	0
3	85	85	0
MI	70	70	0
NY	100	100	0
CA	105	105	0
GA	75	75	0

Objective	
Cost	198200

Figure 4: Excel output for the linear programming problem in Chapter 3, Exercise 15.

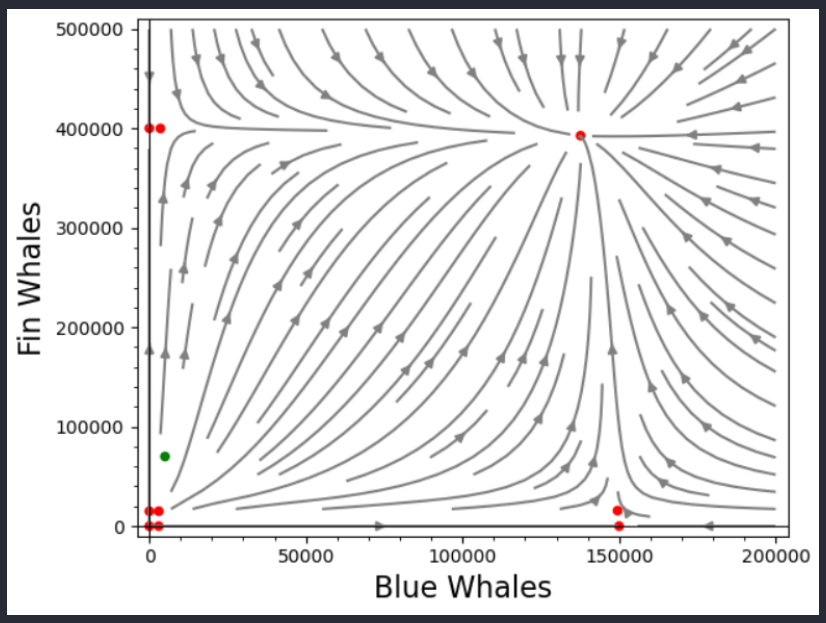


Figure 5: Stream plot of the growth rates of blue and fin whales in Chapter 4, Exercise 5. The red points are equilibriums, each labeled in Table 2. The green point denotes the starting populations for part (c).